# EEG-based on Emotion Recognition Using Machine Learning

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***Abstract*—Emotion is of great significance in human's daily interaction. Contemporarily, emotion recognition systems are beneficial in various areas, ranging from human-computer interaction (HCI) system to healthcare safety. So far, the Electroencephalogram (EEG) has received much attention because of its sensitivity to mood changes. However, the relationship hasn't been fully researched. To investigate the relationship among emotions, bands, channels and find a better classifier, this experiment organize the framework based on the DEAP dataset. After the data preprocessing, EEG data is reduced by principal component analysis (PCA). Then to create the emotion feature vectors, the discrete wavelet transform (DWT) is chosen as the parameter. Lastly, to categorize the EEG signal, support vector machine (SVM), random forest (RF), k-nearest neighbor (KNN), XGboost model is created. According to the experiment, this paper obtained the following relevant conclusions: 1) RF, SVM, KNN and XGboost classifier, performed with the accuracy of 77.01%, 80.44%, 66.75% and 77.75%, respectively; 2) In temperal and whole brain regions, gamma has the highest recognition ability, reaching 76.77% and 77.02%. The beta performs best in occipital, parietal, and central regions with 78.00%, 77.26%, and 78.00%, respectively; 3) The higher the frequency, the greater the oscillatory dynamics of brain activity on positive / negative emotions; 4) The AF3, AF4 area has a greater effect in reflecting emotions because of less irrelevant influences.**

***Keywords—EEG, emotion recognition, DEAP, machine learning***

## I. Introduction

Emotions play a significant role in human interpersonal interaction and human society[1]. Consequently, emotion recognition technology comes into being and develops, widely used in fields of HCI, artificial intelligence,ect. Since emotions significantly affect the human brain's cognition, it can also be utilized in contemporary clinical care, such as depression, post-traumatic stress disorder (PTSD)[2], ect. Therefore, studies on emotion recognition are neccessary. Recently, various resources are used to recognize emotions. Such as facial expressions, speech sounds and electrocardiography(ECG), electromyography(EMG), etc. However, the research shows that speech recognition and multimodal measurements almostly 10 percent less accurate than EEG[3]. Moreover, EEG is easier to collect and cheaper compared with other methods. So, recently, a growing body of researchers have devoted themselves to the EEG emotion recognition study. The relationship between emotion and brain channels and brain electrical channels, as well as the development of the optimal classifier, has become a hot topic.

However, the original EEG signals have properties of interference classification, such as non-stationary, noisy, and high-dimensional[4]. Therefore, it is very important for classifier to filter out EEG features and convert them into numerical features. Nowadays, significant feature extraction and optimal emotion classification are two major parts for EEG-based recognition system[5]. Using machine learning classifiers, emotion recognition had generated good classification results.

Zheng et al[6] took power spectral density, differential entropy (DE), differential asymmetry as extraction features. Finding 1) Waves of different channels reflect different positive and negative emotions. Higher frequency beta and gamma oscillations were more correlated with mood. 2) Temporal lobe location was more correlated with extreme mood. Based on a dimensional model, Atkinson et al[7] extracted the time-frequency features and classified them using kernel-based SVM classifier, and performed with 73.06% accuracy for arousal, 73.14% accuracy for valence. Mohammedi et al [8], selected five frontal pairs of channels to extract energy and entropy of wavelet coefficients. KNN method were applied to classify the emotional states, gained the highest accuracy of 84% and 86% on valence and arousal dimensions. Rahman et al[9] extracted feature by PCA and t-statistics. And he applied SVM and KNN to classify the emotional states, which performances are found 85.85% and 73.42%, respectively. Mehmood et al[10] creates optimal feature selection method and apply RF, four integration methods as classifiers. The highest accuracy of 76.6% for Voting ensembles method. Although some advanced deep learning algorithms may produce excellent classification accuracy, its computational architecture is impractical for the restricted number of classes.

In this paper, our classification model is a two-dimensional classification model of arousal and valence. DWT was used to convert EEG signals into frequency domain signals of four channels. Then frequency domain energy and differential entropy are extracted as features. Next, author utilizes PCA to reduce the EEG features. Four machine-learning algorithms are applied as classifiers. The structure of this study is shown below. Firstly, author represents the details of the dataset, the preprocessing steps and dimensional model. Secondly, this work introduces the details of feature extraction and the methods for feature dimension reduction. Thirdly, author shows the principle of the EEG classifier. Then, author analyzed the contrast experiments and results. Discussion about EEG channels, brain channels, and optimal classifier is conducted. Finally, author give a research summary.



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## II. Material and Methods

A number of steps must be taken from data collection to results, including signal preprocessing, reduction, feature extraction, classifications, etc. This steps have been shown in Fig. 1 and the related details are discussed in this part.

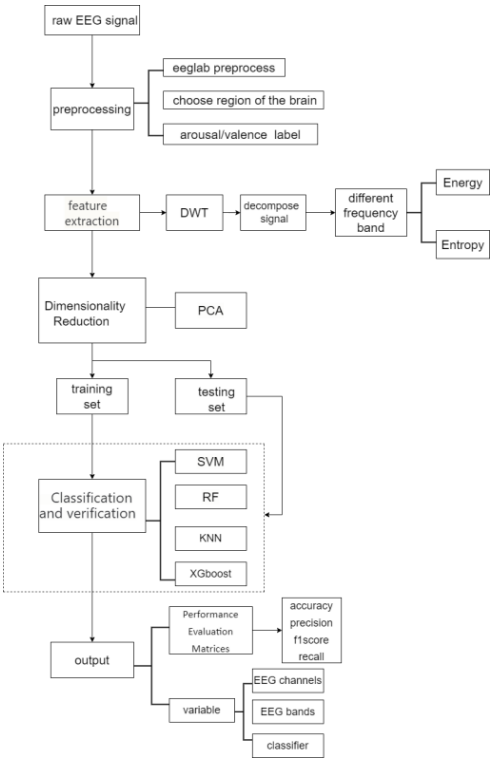


Fig. 1. The framework of EEG emotion recognition method

### A. DEAP database

The experiment was carried on the DEAP dataset developed by a team at Queen Mary University. There were 32 subjects in this dataset, and each subject watched 40 music videos with different emotional orientation, one minute each. During the trial, the EEG was recorded for each participant by the split electrode. The subjects then rated the videos on a scale of 1-9 in terms of arousal, valence,ect. In this paper, signal values from the first 32 brain electrodes were used for the analysis. And, the sample frequency of data is 512 Hz. The channel locations have been shown in Fig. 2, according to the international 10-20 system. After preprocessing, the signal time in each trail is 63 seconds, with the first 3 seconds being the time before the video viewing and the second minute being the video viewing time. The details about the dataset are described in[11].

### B. Preprocessing

The raw EEG data of the DEAP dataset was primarily preprocessed by applying a bandpass frequency filter from 4.0-45.0Hz, and downsampling to 128 Hz. Then, the EEG signal was split into 60s based on the marker, and the first three seconds before the experiment were removed. The removal of ocular artifacts is with the application of ICA. Lastly, the electrical and noise components are eliminated. All of the above treatments were done in the eeglab. After an initial signal processing in the eeglab, the average mean reference method was used. Calculate the average value for each selected channel, and subtracted the mean value from all of the selected channels.

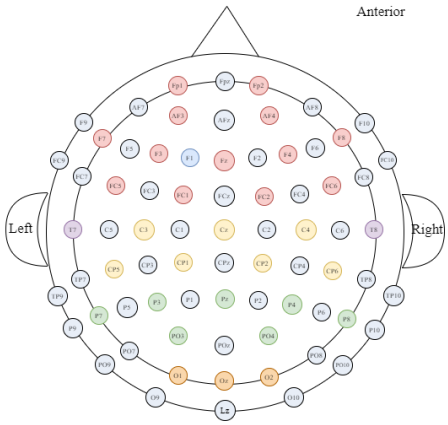


Fig. 2. Electrodes used in the 10-20 international brain positioning system.

Then achieve the purpose of reducing internal and external interference. Meanwhile, the Min-Max normalization was also performed. Through a linear transformation, the samples were transformed under the same dimension. Normalization reduces the impact of individual differences and reduces computational complexity. All the data are mapped to process within a range of 0 to 1.

The processing steps can be compared with the flowsheet given in Fig. 3.

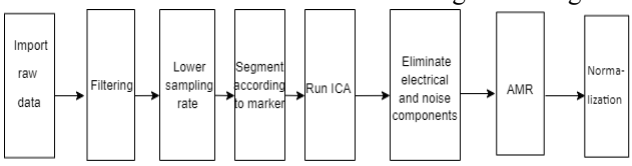


Fig. 3. Processing flowsheet

### C. Emotion Classification models

Dimensional model is expressed in terms of two provoking emotions: Valence and Arousal[6]. Among them, arousal shows the degree of excitement, that is, the intensity of the emotion. The word valence is the degree of satisfaction, both of which are used to divide positive and negative emotions. Since valence and arousal span multidimensional scales, this feature is often used to construct two-dimensional models to represent emotional states. The states are divided into four types: low arousal/low valence, low arousal/high valence, high arousal/low valence and high arousal/high valence. In this experiment, the DEAP dataset defines four emotional states on the arousal and valence dimensions according to 1-9 by subjects, with 4.5 as the high and low dividing line.The coordinate model defined by this 2-dimensional emotion standard is shown in Fig. 4. Each quadrant represents one type of emotion. Different coordinate values in the same quadrant also indicate subtle differences in different emotions.



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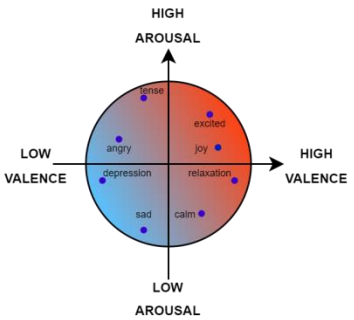


Fig. 4. Emotion types on the valence and arousal scale

*D. Feature extraction*

*1) Wave decomposition*

Studies have shown that different frequency bands of brain waves are correlated with multiple physiological responses, including mental activity, emotional state, cognitive activity, and perceptual function. The alpha, beta, gamma, theta band is often associated with relaxation and sleep, agitation, memory and perception, anxiety and thinking, respectively. In conducting emotion analysis, different brain wave channels, especially high frequency bands, can help us extract useful features to better identify different emotions. Even if some channels are unrelated to emotions, they may provide useful information about the type, intensity and duration of emotions, etc. Therefore, when extracting the EEG features, many scholars will first map the EEG signals to these five frequency bands, and then extract the features corresponding to each frequency band respectively.

In this experiment, wave decomposition was performed by DWT. DWT is characterized by high resolution in non-stationary signals and nonlinear signals. First, the DWT truncates the frequency component of the signal under different scale conditions. Then, the signal passes through high-pass and low-pass filters so that the different frequencies are decomposed. Paying attention to both the number of samples and the degree of overfitting. This time author selected 2s wide and 4s wide windows with coincidence times of 1 and 2s, respectively. A total of 60s signals was divided into 29 and 59 windows. Db4 is the mother wavelet function. Afterwards,the signal time domain information value is converted into the frequency domain information value, it is divided into five channels according to the frequency, including theta (4-8 Hz), alpha (8-16 Hz), beta (16-32 Hz), gamma (32-64 Hz) and noises (> 64 Hz).

For the type of extracted features, because the time-frequency domain features can more accurately extract the emotion features, and the frequency domain features can better understand the data, making the emotion identification more accurate. Therefore, both time-frequency domain features and frequency domain features are essential when performing brain wave analysis. For this analysis, energy and entropy were extracted as features.

*2) Entropy*

DE is a measure of the signal information and can be used to quantify the signal complexity. It is a generalized form of Shannon information entropy on continuous signals. It is calculated by using a probability density function to estimate the probability of each frequency bin in the signal. DE can be used for comparing signals across different domains and detecting changes in the signal over time. DE is a powerful tool for feature extraction and can be used to improve the model performance.

In this trial, author used the EEG values for each channel to calculate the DE as extracted features. The entropy is calculated by summing the squares of the values in the channel multiplied by their natural logarithm. The DE formula for each channel is computed as:

$$ENT_i = -\sum_{k=1}^{N}(D_i(k)^2)log(D_i(k)^2) \quad (1)$$

The i represents the level of wavelet decomposition in the band, and the k represents the number of wavelet coefficients within the i band.

*3) Energy*

Frequency domain energy can reflect the information of brain waves on different channels, and each frequency energy has a different meaning. When a person is feeling excited, the EEG usually shows an increase in gamma waves and a decrease in alpha. In the opposite case, the EEG usually shows a decrease in gamma and an increase in theta. Thus, by analyzing the frequency-domain energy, it helps to identify emotions more accurately. In this experiment, the energies of the different channels were extracted for the analysis. The frequency domain energy is defined as:

$$E(x) = \sum_n |x(n)|^2 \quad (2)$$

where x(n) is the signal of the brain wave on the frequency n.

*E. Dimensionality Reduction*

PCA has been applied to reduce the dimensions of the extracted EEG features. PCA can reduce the number of features by fusing multiple variables into new variables. The optimization goal of PCA is to lower the higher order N dimensional vectors to lower order K dimensions. By selecting the orthogonal basis of K, the original data is transformed with the covariance of 0 and the variance is as large as possible. Using the diagonizable properties of the real symmetric matrix, this paper can obtain the data matrix with dimension reduction. The main step is as follows,

$$E^T C E = \Lambda = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix} \quad (3)$$

*F. Classification of Emotions*

After extracting the EEG features, author choses several supervised learning emotion classifier, which are specifically described as follows.

*1) SVM*

The SVM is a supervised and nonlinear binary classifier. SVM uses the kernel function and can be transformed from low to high dimensions. So the separated hyperplane that divides the training dataset with the largest geometric interval could be found. At the same time, the fault tolerance rate can also be controlled by adjusting the parameters.

The output of a binary SVM classifier can be expressed as :



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$$y = sgn(\sum_{i=1}^{N} \alpha_i y_i k(x_i, x) + b) \quad (4)$$

Gaussian kernel function was used in this experiment,

$$k(x_i, x_j) = exp\left(\frac{-\|x_i - x_j\|^2}{2\delta^2}\right) \quad (5)$$

*2) RF*

RF utilizes the decision tree learning principle. The technique creates a large number of decision trees. Each decision tree is trained on a different subset. The decisions made by the trees are then combined to create a final prediction. In this experiment, GridSearchCV was used to cross-verify the parameters and select the optimal model.

*3) XGboost*

The XBboost is an implementation of the boosting algorithm, and the main effect is to reduce the bias. The objective function includes the model error and the structural error of the model. Better trees are determined by the objective function. Through this algorithm, new trees are constantly generated. Each tree reduces the model bias by learning the difference between the previous tree and the target value. The results of all the trees add up to the predicted value of the model for a sample.

*4) KNN*

The core of the kNN algorithm is that each sample can be represented by its closest k neighbors. The specific steps are shown as follows:

a) Calculate the distance between the objects to be classified and other objects. The distance is calculated as

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \ldots + (x_n - y_n)^2} = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \quad (6)$$

b) Count the nearest K samples.

c) Account the frequency of the category where the K points are located.

d) The category with the highest return frequency was used as the predictive classification.

### G. Performance Evaluation Matrices

*1) Accuracy.* Accuracy is the most basic evaluation indicator, which refers to the percentage of correct predictions made by our model. In this classification problem, he represents the number of correct predictions made by the model for predicting the emotion type. The Accuracy is calculated as

$$accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (7)$$

The meaning of parameter is shown in the table I.

TABLE I. PARAMETER MEANING

| | 1 | 0 |
|---|---|---|
| 1 | True Positives(TP) | False Negative(FN) |
| 0 | False Positive(FP) | True Negative(TN) |

*2) Recall.* It's also known as the TPR curve of sensitivity. Recall describes how many of the true positive cases in the test set are picked out. The Recall is calculated as

$$recall = \frac{TP}{TP+FN} \quad (8)$$

*3) Precision.* Precision describes the number of true positive cases in the positive cases predicted by the binary classifier. The Precision is calculated as

$$precision = \frac{TP}{TP+FP} \quad (9)$$

*4) F1 Score.* Precision and Recall are usually a pair of contradictory evaluation indicators. For a more comprehensive evaluation of the emotion classification model, and consider both accuracy and recall, author used F1 for the evaluation. The F1 score is calculated as

$$F1score = \frac{2*precision*recall}{recall+precision} \quad (10)$$

## III. EXPERIMENT AND RESULTS

### A. Different brain locations and channels

Figure 5 and table II show the accuracy of RF in 5 frequency bands in different brain locations. The results show that the emotion recognition ability of each channel and brain part are different. By analyzing the line map, gamma had the highest recognition ability of 76.77% and 77.02%. And the beta in occipital, parietal, and central regions with 78.00%, 77.26%, and 78.00%, respectively. In frontal region, the recognition level of each frequency band was relatively similar.

TABLE II. THE RF ACCURACY RATES(%) FOR DIFFERENT LOCATIONS OBTAINED FROM SEPARATE AND TOTAL

| bands | label | Frontal lobe | Temporal lobes | Occipital lobes | Parietal lobes | Central | Total |
|---|---|---|---|---|---|---|---|
| Theta | Arousal | 75. 55 | 75. 06 | 76. 53 | 75. 55 | 77. 26 | 74. 08 |
| | valence | 69. 55 | 67. 08 | 72. 03 | 71. 04 | 71. 78 | 69. 55 |
| Alpha | Arousal | 75. 06 | 75. 06 | 76. 77 | 72. 86 | 75. 31 | 75. 55 |
| | valence | 66. 83 | 71. 78 | 69. 55 | 69. 80 | 71. 78 | 69. 31 |
| Beta | Arousal | 75. 55 | 73. 83 | 78. 00 | 77. 26 | 78. 00 | 74. 08 |
| | valence | 73. 27 | 72. 28 | 72. 28 | 71. 53 | 74. 01 | 70. 79 |
| Gamma | Arousal | 74. 08 | 76. 77 | 73. 35 | 74. 08 | 75. 79 | 77. 02 |
| | valence | 74. 26 | 68. 81 | 75. 50 | 72. 77 | 74. 01 | 73. 27 |
| Total | Arousal | 72. 62 | 71. 00 | 73. 96 | 72. 31 | 73. 35 | 72. 01 |
| | valence | 71. 41 | 69. 86 | 71. 78 | 71. 35 | 72. 77 | 73. 02 |



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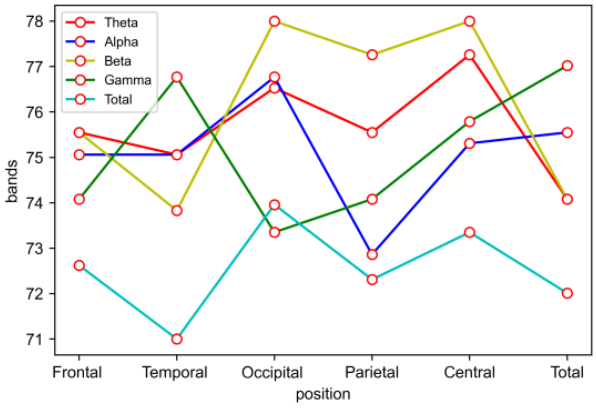


Fig. 5. Accuracy performance of different channels in different parts

## B. Different emotions

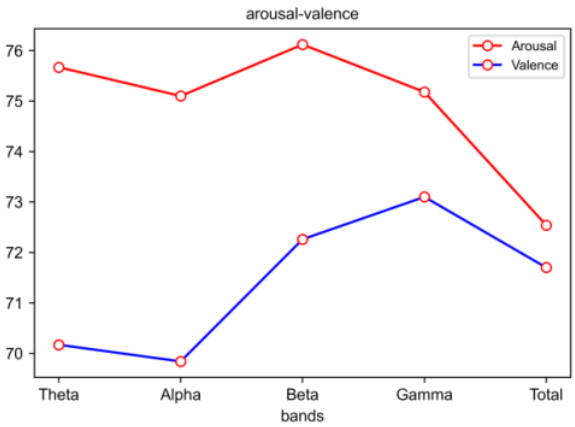


Fig. 6. arousal-valence in different bands

Figure 6 shows the difference in RF accuracy of different frequency bands in performance arousal and valence. By observing the graph lines, the excitement recognition accuracy is higher than the satisfaction degree. It indicates that the arousal showing the degree of excitement reflects the brain wave band more violently and is easier to distinguish. Overall, beta and gamma had higher average recognition accuracy for excitement and happy satisfaction. This suggests that the higher the frequency, the greater the oscillatory dynamics of brain activity contributes to positive / negative emotions.

## C. Asymmetric channel

Figure 7 and table III show the accuracy performance of different frequency bands in prefrontal and frontal regions. As can be seen from the figure, both frontal and frontal regions performed better for mood. However, the forehead area has a greater role in reflecting emotions because it receives more external influences.

TABLE III. RF AROUSAL ACCURACY RATES(%) FROM EACH PAIR OF FRONT CHANNELS IN DIFFERENT BANDS

| bands | Fp1 Fp2 | AF3 AF4 | F3 F4 | F7 F8 | FC5 FC6 | FC1 FC2 |
|---|---|---|---|---|---|---|
| Theta | 74.57 | 77.26 | 71.64 | 75.31 | 73.35 | 74.82 |
| Alpha | 75.06 | 76.28 | 74.82 | 72.86 | 75.79 | 72.62 |
| Beta | 70.90 | 73.59 | 72.37 | 74.33 | 75.79 | 73.12 |
| Gamma | 72.62 | 78.73 | 73.35 | 75.79 | 71.64 | 71.88 |

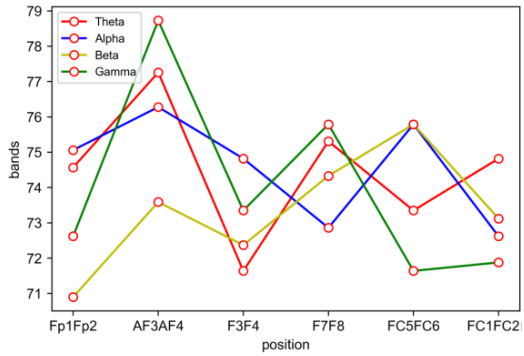


Fig. 7. Performance of the asymmetric electrodes in different frequency bands

## D. Different machine learning classification

TABLE IV. MACHINE LEARNING CLASSIFICATION AVERAGE RESULTS COMPARISON

| classifier | RF | SVM | KNN | Xgboost |
|---|---|---|---|---|
| accuracy | 77. 01% | 80. 44% | 66. 75% | 77. 75% |
| recall | 0. 81 | 0. 64 | 0. 71 | 0. 81 |
| precision | 0. 77 | 0. 78 | 0. 68 | 0. 78 |
| F1score | 0. 79 | 0. 78 | 0. 70 | 0. 80 |
| roc | 0. 83 | 0. 82 | 0. 73 | 0. 85 |

Table IV compared the performance of four different classifiers in the arousal labels, KNN, RF, SVM, and Xgboost in all channels of the same channel. In this assessment, the DE and energy features were used as input to the classifier. KNN, with a parameter K of 3. For RF and linear SVM, a cross-validated grid search was used to adjust the parameters. The indicators of RF, KNN, Xgboost, and RBF core SVM are shown in Table IV.

# IV. DISCUSSION

## A. Beta wave in central electrode

The confusion matrix and ROC curve in the central part of beta wave were analyzed (see Figure 8 & 9). Using the relevant knowledge of neuroscience to explain the high rate of emotion recognition in this channel and location.

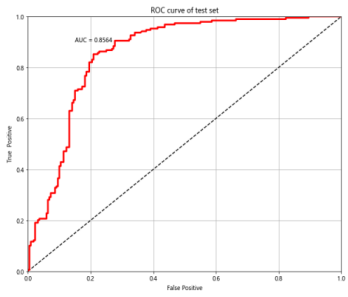


Fig. 8. ROC curve of the beta in central

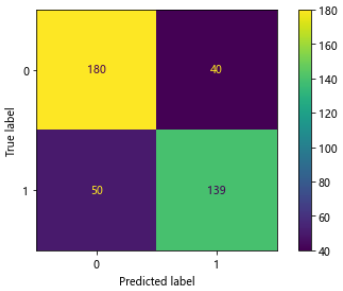


Fig. 9. Confusion matrix of the beta in central



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Beta is a fast wave, often seen in a state of clarity. High levels of beta have stimulation, arousal, and alertness. When nervous, emotional, or excited, they produce a beta wave, which is often used to show the arousal of the brain. When the activity of the fast composition of the ripple was enhanced, the state of the system was reduced, the central nervous system suppressed the state relief, increased the degree of alertness, the alertness and the increased excitement. In this experiment, the beta-wave activity was enhanced when the participants received the stimulus, indicating an increased emotional excitability. Thus, the arousal arousal value rises and is more easily recognized by the classifier.

### *B. Gamma wave in AF3, AF4 electrode*

The confusion matrix and the ROC curve on AF3, AF4 were analyzed (see Figure 10 & 11). The relevant knowledge of neuroscience is used to explain the high emotion recognition rate.

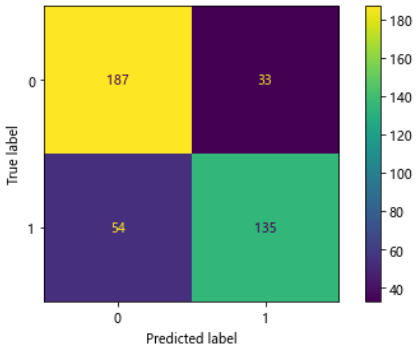


Fig. 10. Confusion matrix of the gamma in AF3, AF4

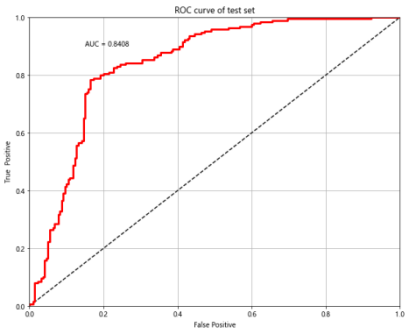


Fig. 11. ROC curve of the gamma in AF3,AF4

The Gamma wave is a high frequency of neuronal activity that can be produced in different regions of the brain and provides an effective way to measure the activity level of the brain. Studies have shown that gamma wave intensity is associated with emotional tension. When people are in emotional stress, the gamma wave intensity increases, which may result in higher arousal values. Similarly, the gamma wave intensity decreases when people are in emotional calm, which may also lead to a decrease in arousal values.

AF3 and AF4 are two channels in the EEG that can be used to monitor electrical activity in the brain and can measure the degree of emotional agitation. It was shown that the gamma wave intensity increases when the two channels are stimulated, thus also increasing the arousal value. Therefore, AF3 and AF4 can be used to directly measure the degree of emotional excitement, and can indirectly measure the high and low arousal values. AF3 and AF4 are more reflective of emotional agitation values than other frontal channels. Because they are located in the frontal region, this region can more effectively reflect the degree of emotional agitation without being influenced by other factors like other brain channels.

## V. Conclusion

In this study, the EEG signals were decomposed into 5-frequency bands using publicly available DEAP data sets. Next, the energy and input features are fed to the classifier. Using RF, SVM, KNN, and xgboost classifiers, this paper classified the emotional states with an accuracy of 77. 01%, 80.44%, 66.75%, and 77.75%, respectively. Through multiple groups of experiments, author found that gamma had the highest recognition ability, reaching 76.77% and 77.02%. And beta in occipital, parietal and central, reaching 78.00%, 77.26% and 78.00%, respectively. In frontal region, the recognition level was similar. The high average recognition accuracy of beta and gamma for excitement and happy satisfaction indicates that the higher the frequency, the greater the oscillatory dynamics of brain activity is to positive/negative emotions. The frontal region plays a bigger role in reflecting emotions because it is more affected by emotional stimuli and less affected by irrelevant ones.

In the current work, we mainly studied the dichotomous status of two positive indicators, and there will be positive, neutral and negative diversified classifications in the future. In terms of features, the relevant principles of neuroscience will be combined with the method of simulation modeling. More features, such as rational asymmetry and differential cauality, will also be added to make the prediction basis more comprehensive. In terms of classifiers, we will add simulation modeling, deep learning methods in the future, or adopt a new combination of multiple classifiers to achieve higher cross-validation accuracy.

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